Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Systems Fundamentals I & II, Algorithms

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations with enforced ban/edit permissions to prevent client-side database tampering, and authentication features including a password reset flow and custom claims for new-user onboarding.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, and custom date/time picker integration to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, a music-tutoring scheduling app, managing schema migrations via Supabase CLI and configuring auth/login flows; built a seed-data workflow that let the team build and test independently.

PROJECTS

Diary Handwriting Analysis Python, PyTorch, Hugging Face Transformers, OpenCV

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline, reaching ~87% reliable page transcription; extending to DANIEL, a segmentation-free architecture for joint transcription and named-entity extraction to surface mood and topic patterns across diary entries over time.

TuneAcademy React, NumPy, Essentia, Firestore

- Built the AI/audio-analysis pipeline for an app matching music students with instructors, using Essentia and NumPy to extract pitch, rhythm, and tone features and rule-based scoring algorithms to detect technical weaknesses within a 36-hour hackathon build window.
- Set up a CI/CD pipeline with GitHub Actions to lint, test, and auto-deploy the app to Cloudflare Workers, backed by a React/Tailwind frontend, Python/FastAPI backend, and Cloud Firestore database.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University Sep 2025 – Dec 2025

- Built a pipeline using Selenium and Python to monitor newly issued SSL/TLS certificates in real time and flag fraudulent domains impersonating EZ-Pass tolling services.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original midterm review materials adopted as the basis for later semesters and hosted review sessions reaching 100+ students.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, Firebase/Firestore, MongoDB, Supabase, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium